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Text Interviews on Mobile Devices

Frederick G. Conrad,^{1,2} Michael F. Schober,³ Christopher Antoun,^{*,4} Andrew L. Hupp,¹ and H. Yanna Yan¹

¹ Survey Research Center, University of Michigan, Ann Arbor, MI, USA

² Joint Program in Survey Methodology, University of Maryland, College Park, MD, USA

³ Department of Psychology, New School for Social Research, New York, NY, USA

⁴ Center for Survey Measurement, U.S. Census Bureau, Suitland, MD, USA

People's communicative habits are changing with the rise of mobile devices. Increasingly, people are communicating when, where and in ways that are convenient and practical for the task at hand: talking in real time (with or without video), exchanging recorded spoken messages, exchanging written messages (through email or text messaging, or on social media sites), and in more than one mode at a time (e.g., integrating still images, audio or video with textual messages, or textually annotating an image they are sending). Even the most basic mobile phones, which in some parts of the world account for the majority of use (Poushter et al., 2015), allow text messaging. For an increasing percentage of the world's population, texting is a primary method of daily communication, whether it is on a multimodal mobile device (e.g., a smartphone or tablet) or a phone with fewer multimodal options, and whether the text message travels via internet or a cellular network.

The daily prevalence of text messaging (e.g., 156.7 billion text messages sent per month in the U.S. by the end of 2015, CTIA, 2016) makes it a mode worth exploring for survey data collection—not only for recruiting participants into an online or telephone survey but also as the primary mode of asking survey questions and collecting answers. A survey administered via texting blurs the boundaries of what should be considered an “interview”: texting has a back-and-forth turn-taking structure like a spoken conversation but less demand for an immediate response and little evidence of who is on the other end, making it more like a “self-administered” online interaction. Surveys are already sometimes administered via texting, particularly often in parts of the world that do not have well-developed technological or administrative infrastructures for social measurement (e.g., Johnson, 2016), and globally the number of potential text interview participants is large (and increasing). There may even be subgroups of populations who may only be successfully contacted by text messaging (i.e., they simply do not

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engage in email or phone communication). But the properties of text interviews are not well understood.

As we see it, much remains to be learned about the quality of the data produced by text interviews and the viability of the mode. Our aim in this chapter is to summarize what is currently known about collecting survey data via text messaging, with a particular focus on coverage, non-response, and measurement error. We first discuss basic properties of text communication and then turn to texting as a tool for researchers who conduct surveys.

14.1 Texting as a Way of Interacting

14.1.1 Properties and Affordances

When people text, some aspects of their experience are much the same irrespective of the devices they use—that is, some properties of texting are stable. Other aspects vary depending on the devices and networks involved. In text survey interviews, these stable and variable features are particularly relevant to *respondents'* experience; *interviewers*, in contrast, might well send and receive texts through an entirely different interface than respondents and probably not on a mobile device (e.g., from a desktop computer), depending on the design of the particular text survey. Of course, automated interviewing systems do not involve an interviewer interface at all.

14.1.1.1 Stable Properties

Texting is composed of a set of features that distinguish it from other modes of interaction (Thurlow and Poff, 2013) no matter what device or network is used for sending and receiving the texts. A comparison of the features of text and voice interaction on mobile devices highlights what is unique about texting (see Table 14.1).

Texting is analogous to spoken conversation on a number of fronts: for example, both have a back-and-forth turn-taking structure, and in both cases one contacts communication partners through a mobile phone number. But texting is inherently asynchronous, in that the back-and-forth of messages does not have to occur in real time: there can be delays of days or weeks between the sending of messages within a particular conversation. Even when the exchange of texts is rapid, what is exchanged are full “packages” of message content that appear all at once, unlike speech, which unfolds sound by sound over time. In contrast, voice communication is inherently synchronous, and too-long pauses between speaking turns (longer than about a second, Jefferson, 1988; Roberts and Francis, 2013) are interpreted as meaningful in different ways than text delays are interpreted as meaningful. That is, speech delays are seen as evidence that the upcoming speaker is having trouble formulating or finding the right words for a next utterance, or that the speaker’s attention has lapsed, or that the speaker feels awkward, and so on (Goldman-Eisler, 1958). In texting, in contrast, long response times may not indicate trouble at all; a delay (in a situation that demands a response) could indicate that the upcoming texter has not even read the message, or is otherwise occupied, or wants to think about their response.

Texting leaves a visible trace of at least the most recent communication (i.e., it is “reviewable,” to use Clark and Brennan’s (1991) term), and on a smartphone can be displayed as a fully threaded record of the conversation that can be reread at will. In contrast, voice is ephemeral, and speakers and listeners must rely on their memories to know what was said earlier (unless they have access to audio recordings). Usually, text messages are private—bystanders are unlikely to be able to see what has been texted—while in voice conversations a bystander

Table 14.1 Properties of mobile voice and text communication, originally appeared in Schober et al. (2015).

Property	Voice	Text
Synchrony	Fully synchronous	Less or asynchronous
Medium	Auditory	Visual
Language	Spoken/heard	Written/read
Conversational structure	Turn-by-turn, with potential for simultaneous speech	Turn-by-turn, rarely but possibly out-of-sequence
Persistence of turn	No	Yes
Persistence of entire conversation	No	Yes; on smartphones, threaded and fully visible
Social presence of partner	Continuous (auditory) presence	Intermittent evidence (when texts arrive)
Nonverbal cues of emotional state and intentions	Always present: speech always has intonation and stress contours, as well as timing (e.g., pauses)	Only present if added by sender through words (e.g., “LOL”), orthography (e.g., capital letters, punctuation) or emoticons (e.g., ☺)
Character of multitasking	Simultaneous, especially when hands free, unless other task involves talking	Switching required between texting and other tasks
Impact of environmental conditions	Potential interference from ambient noise	Potential interference from visual glare
Impact of nearby others	Others may hear conversation; potential audio interference from others’ talk	Others unlikely to see content on screen, though possible
Required connection to network (cellular or Wi-Fi)	Must be continuous	Can be intermittent

may have full access to at least one side of the conversation (Monk et al., 2004). The asynchrony of texting and the visibility of the record of the conversation allow texters to carry out other activities at the same time as they are engaged in ongoing text conversations differently than speakers can. People can respond to texts at a time and place that is convenient for them, momentarily switching to the text from the other tasks they are carrying out, while successful voice communication demands ongoing attention. Text interactions can be unaffected by intermittent network (cellular or Wi-Fi) connections as long as the connection is sufficiently strong at the moments of sending and receiving, while voice calls require continued connectivity.

14.1.1.2 Properties That Vary across Devices and Networks

Other features of texting vary depending on the particular communication devices and communication networks involved. Devices from different eras—which make different bundles of features available—are simultaneously in use, and in different proportions in different areas of the world (Poushter et al., 2015). As devices have changed, so have the services through which text messages are exchanged. SMS (short message service) technology, in which short alphanumeric text-based messages are sent via the spare capacity of a digital cellular telephone network (Taylor and Vincent, 2005), is the most basic and universal way to text, but its costs vary from provider to provider and from country to country, and pricing models continue to change (e.g., per message vs. unlimited plans). With the arrival of internet-enabled mobile devices like smartphones and tablets, lower- or even no-cost alternatives to SMS (e.g., mobile apps such as

WhatsApp and messaging within social media sites such as Facebook) have become prevalent. This means that researchers interested in interviewing via text will need to be especially sensitive to the distribution of different devices and services at any one moment within the population of interest.

Character limits. The most basic version of texting limits the number of characters that can be sent at one time to 160 alphanumeric characters, or 70 characters in non-Latin-alphabet languages. This is a result of the constraints of SMS technology. As messaging services have grown to include internet-based networks and more sophisticated messaging apps, the character limits have also changed: depending on the particular texting service there may be no maximum character length for text messages. Nonetheless, it is likely that at least some members of a population of interest will have character limits on their texting.

Input interface. A major difference between different mobile devices is how users enter characters for text messaging. On basic phones, text messages are entered via the (physical) numeric keypad, such that selecting particular letters to enter into a text message requires pressing the numeric keys variable numbers of times (e.g., the characters “g,” “h,” and “i” are all accessed via pressing the number “4” from one to three times: once for “g,” twice for “h,” and three times for “i”) and to text a number requires pressing a number key four times to cycle through the associated letters. Another input interface is the full physical alphanumeric keyboard, which allows additional symbols to be texted. A third major input interface is the virtual keyboard, which makes available a broader range of potential character sets (e.g., from many different alphabets and symbol systems) and graphical content (e.g., emojis). Virtual keyboards can also adapt to the orientation of the screen, which allows for easier typing and a far less bulky device. A fourth input possibility, at least on some devices, is spoken language: dictating a text message that a speech recognition system decodes and enters as text. For now, text survey interviews of a given population are likely to involve at least some participants using each of these interfaces.

A further complexity is that in some cases, the same text interaction can be displayed and contributed to on multiple devices, each with a different input interface. For example, a user who has multiple interoperable devices might start an interaction on her mobile phone, continue it on her desktop computer, and complete it on her tablet. If a desktop computer is involved, then an entirely different input interface (keyboard and mouse, along with potential access to the full suite of other software and content available on the desktop) can be part of the picture as well. This means that a researcher implementing a text survey cannot necessarily assume that all respondents continue to use a single device throughout a text interview, which may affect their input capabilities differently at different moments.

Display of texts. Mobile devices differ in how they display the record of text interaction. More basic (“feature”) phones function like email, with an inbox of inbound messages and a “Sent” folder of outgoing messages; different phones vary in the particulars, but usually only one message can be seen at a time, and messages are not grouped by a particular conversation. In contrast, the norm for smartphones is message threading, in which a user can see both sides of a particular conversation on one screen, arrayed in chronological order. For text survey interviews, the threading feature makes the full record of an interview available—all the previous questions and all the previous responses—while for nonthreaded interfaces, respondents do not have this full history immediately available.

Nature of alerts. Users may receive an attention-demanding alert every time a message arrives depending on the device capabilities and how a user has set up preferences for receiving alerts (a tone, a vibration, a visual display) for incoming text messages. In a text interview, this means that respondents with alerts will be re-engaged by the device with every incoming question, while those with no alerts will need to take the initiative to continue the interview.

Subscription-based features. The features available to a particular user—e.g., the ability to attach audio or visual material, or to go beyond a 160-character limit—can vary depending on which application or service they use to send and receive messages. In some messaging apps, both parties must be subscribers to the service to contact each other at all (e.g., messaging within Facebook or WhatsApp is only possible with other Facebook or WhatsApp users) and to use the full range of features. For text interviews, this affects both the sampling frame (e.g., all WhatsApp users or everyone with a mobile device) and the format of what can be presented to a respondent (e.g., visual images or not).

14.2 Contacting and Inviting Potential Respondents through Text

Although our focus here is on texting as an interviewing mode (for asking questions and recording answers), texting is also increasingly used as a method of contact and recruitment (prenotification, invitation, reminder) for potential respondents in other interviewing modes: online, both desktop (Bosnjak et al., 2008) and mobile (de Bruijne and Wijnant, 2014; Marlar et al., 2014; Mavletova and Couper, 2014; Steeh and Piekarski, 2008), and for mixed-mode data collection (Holland et al., 2014). It can also be part of a multimodal suite of recruitment, with a follow-up text message to panelists already recruited through a different mode alerting them to a subsequent interview via text (Down and Duke, 2003).

Invitations to participate in a survey via text have certain attractive features. Text invitations are persistent: they remain in the text inbox or in the text messaging app to be read (or reread) at any time. This all but eliminates the possibility of noncontact (unless the sample member loses or does not consult his or her mobile device), which in voice recruitment occurs when potential respondents do not answer their phones.

Thus far, the evidence suggests that contacting potential respondents through text can be effective and may have benefits in particular populations. For example, Holland et al. (2014) examined the effectiveness of using text messaging as a prenotification mode with nearly 2,500 soldiers in the U.S. Army as part of the Army Study to Assess Risk and Resilience in Servicemembers. Soldiers were randomly assigned to one of four prenotification conditions: no prenotification, letter, text message, or letter and text message. After the prenotification, they received an email invitation with a link to complete a web survey. Soldiers who were prenotified by text message (alone or along with a letter) were more likely to complete the questionnaire, without any reminders, than were soldiers who were prenotified only by a letter or not prenotified. The main advantage of text prenotification was the speed of response to the email invitation. Once the researchers reminded respondents (by email and phone), the advantages of the text prenotification were diminished. At least among this young and highly mobile population, text message prenotifications seem effective during the recruitment phase of a survey, and as Holland et al. propose may shorten the data collection field period. Text invitations may also be effective in increasing mobile web response rates, see Chapter 7.

14.3 Texting as an Interview Mode

Our focus here is on the most direct application of text messaging to interviewing, namely presenting survey questions to respondents via text messages, with respondents answering via text messages. Early feasibility tests (Cooke et al., 2003; Down and Duke, 2003) concluded that the mode was promising in producing a high percent of completed cases in a short time, and might

be particularly useful for certain demographic segments; but text interviews produced different distributions of responses for some questions compared to an online benchmark survey (Down and Duke, 2003). Whether this is because of coverage, nonresponse, or measurement error is unclear. The conclusion at the time was that text interviewing was more likely to be an auxiliary rather than a primary mode of data collection—just one tool in a researcher’s tool chest. But things have changed since then; now that mobile phone use and texting have become so much more pervasive in daily communication, interviewing via texting may be worth considering more seriously as a mainstream mode of data collection.

One potential benefit of interviewing via text messaging is that it lends itself to collecting survey responses close in time to the behaviors they describe more easily than do other survey modes, in what has been called “experience sampling” and “ecological momentary assessment.” For example, with text messaging, one can ask survey respondents to (briefly) report on what they are doing or thinking at the moment they receive the message, and this can potentially be repeated at many times during the data collection period—even daily (Anhøj and Møldrup, 2004; Berkman et al., 2011; Brenner and DeLamater, 2013, 2014; Garcia et al., 2014; Moreno et al., 2012). This would likely be intrusive and impractical with a voice contact, at least by a human interviewer; respondents are unlikely to answer their phone several times a day (possibly at random moments) to answer one question, while they may be willing to respond to texted questions on such a schedule. Concurrent reporting would also be impractical with a desktop web survey, given that respondents’ frequent mobility makes it unlikely they would be able to respond in that mode when prompted. (A *mobile* web survey in which regular detectable alerts are sent would begin to approximate a text messaging interview.) The fact that this kind of data collection can happen via text messaging makes concurrent reporting via text as viable as retrospective reporting.

A related—though not identical—application of text messaging that can involve data collection is behavioral intervention, in which researchers or medical professionals interact with patients via text messaging both to send information and reminders (e.g., about taking medicine or exercising) and also to collect data from the patients (e.g., verifying that they have taken medicine, or reporting a current value on a physiological measure like heart rate) (see Fjeldsoe et al. (2009) for one meta-analysis). Text messaging has been used for behavioral interventions in disease prevention and management (Cole-Lewis and Kershaw, 2010), smoking cessation programs (Ybarra et al., 2014), weight loss (Patrick et al., 2009), antiretroviral therapy (Pop-Eleches et al., 2011), and reducing alcohol consumption (Suffoletto et al., 2012), among many others. The data collection in these domains is different from conventional surveys in that it is designed to evaluate the intervention rather than to produce population estimates, but there may well be lessons from behavioral intervention texting that could inform the design of text survey interviews.

If text interviewing is to be added to the survey research toolkit, researchers must grapple with issues relevant to total survey error (TSE) that are particular to the mode. From a TSE perspective, this means being able to characterize (and minimize) coverage error, sampling error, non-response error, and measurement error, as well as the efficiency of text interviewing relative to other modes. We next review what is known thus far about these.

14.3.1 Coverage and Sampling Error

If coverage issues limit who can be reached for a text interview, estimates derived from these interviews will be biased. How might one select a representative sample of text users?

Unlike traditional web surveys, for which national sampling frames do not exist, it is conceptually straightforward to generate probability-based samples of mobile phone numbers through

random-digit-dial (RDD) procedures;¹ because text capability is native to virtually all mobile phones, the sampling issues for text and voice interviews should be identical: one can ask spoken questions to and receive spoken answers from the users associated with the same mobile phone numbers to which one can text questions and from which one can receive texted answers. This approach necessarily excludes landline numbers, but any bias this might introduce is likely to become less problematic as landline phones are replaced by mobiles, so-called “wireless substitution” (Blumberg and Luke, 2011), and because in some cases cell-only samples are more efficient than dual-frame designs for sampling both landline and mobile-only users (Peytchev and Neely, 2013). A larger practical challenge arises from the legal prohibition against sending unsolicited text messages (Link et al., 2014), at least in the U.S. In places with such restrictions, researchers will need to gain consent from sample members through other means, e.g., a voice call (perhaps in a prior wave of a longitudinal study), before sending them a text invitation.

While samples of mobile phone users may increasingly represent the larger population, it is not known to what extent text users represent the population of mobile phone users. Although mobile phones make SMS—and other text apps on smartphones—available to all users, some mobile phone users do not text. The undercoverage of—actually the exclusion of—mobile phone owners by those who text may bias estimates. The amount of undercoverage error depends both on the proportion of the target population who use text messaging and on the differences between texters and nontexters. The first issue is that those mobile phone owners who do not send and receive texts can never participate in a text interview. It is estimated that texting use among mobile owners is about 70% in Europe and 81% in the U.S. (Duggan, 2013; Kohut et al., 2012), although, of course, these proportions may well change—probably increase—over time. The other issue is that those who text may differ from those who do not. As of 2003, when texting was less ubiquitous and before the rise of smartphones, older members of the U.K. population seemed to be substantially under-represented (Down and Duke, 2003). More recently, researchers at the Pew Research Center have found that texters are younger, better educated, and more likely to be Hispanic compared to nontexting mobile owners (Duggan and Rainie, 2012). How the characteristics of texters will evolve over time remains unknown.

Few studies have explored whether coverage error might affect the accuracy of estimates based on text surveys. Here, we begin to explore this issue by examining one national population (U.S.) using data from one year (2013), simulating who would be covered in a national cell-RDD text survey. Using data from a telephone survey conducted in the U.S. by Pew Research Center (2013), we compare sociodemographic characteristics from the benchmark population (the full mobile sample) and the population of texters (mobile owners who report using their phones to text), and then estimate coverage bias for each statistic (Table 14.2).

In this case, most estimates of texters’ characteristics match the full sample values well; only the estimates for age and education diverge significantly from the full population estimates. There are fewer older adults among the texters than in the full sample, and there are more college graduates among the texters than in the full sample. Beyond demographic biases, substantive variables—for example, measures of health—could also be biased in text surveys if the demographic variables related to mobile phone ownership and texting are also related to health (or other variables of interest). This may be addressed through weighting if the researcher knows which demographic variables are related to the measure of interest (e.g., age tends to correlate with health, and so if younger users are over-represented in a text sample this could lead to underestimates of disease); but weighting can solve this problem only if the mediating variable is identified and measured.

1 Note that it is also possible to recruit participants into mobile web surveys by sampling mobile phone numbers (see Chapter 7).

Table 14.2 Absolute coverage bias estimates of texter population relative to U.S. mobile phone owners (2013) for six sociodemographic statistics.

	Full sample of mobile owners (<i>n</i> = 1,127) % (SE)	Mobile owners who text (<i>n</i> = 962) % (SE)	Coverage deviation ^{a)} Est. (SE) ^{b)}
Age (>34 years old)	62.4 (1.4)	57.0 (1.6)	5.4* (0.4)
Gender (=male)	56.2 (1.5)	55.3 (1.6)	0.9 (0.6)
Education (=college grad)	36.4 (1.4)	38.0 (1.6)	1.7* (0.6)
Race (=white)	70.5 (1.4)	70.1 (1.6)	0.5 (0.5)
Ethnicity (=Hispanic)	16.1 (1.1)	16.4 (1.2)	0.4 (0.4)
Income (=\$30K or more)	71.8 (1.3)	72.9 (1.4)	1.1 (0.6)

Data Source: Pew’s Spring Tracking Survey conducted during April–May 2013.

- a) $|y(\text{cell owners who text}) - y(\text{full sample})|$.
b) Calculated using a formula described by Lee (2006, p. 465).
* $p < 0.05$.

To the extent that there is coverage error, this must be considered in light of increasing penetration of mobile ownership and texting, i.e., as the percentage of mobile owners who are not text users decreases, the impact of any differences between texters and nontexting mobile owners will decrease as well. As shown in Figure 14.1, texting rates among mobile owners grew notably between 2007 and 2013, increasing from 58 to 81% (Pew Research Center, 2013). We fit a linear trend line and it is highly correlated with actual growth ($r^2 = 0.95$). Texting may never become universal: there may always be some who will not have a mobile phone at all, and there will be mobile phone owners who are unable or unwilling to text due to motor and visual impairments or poor reading skills (although text-to-speech advances may enable some members of these populations to use text as well). In any case, if current trends continue then most cell owners are likely to become text users (in one format or another), so that coverage error may not be a concern for most topics.

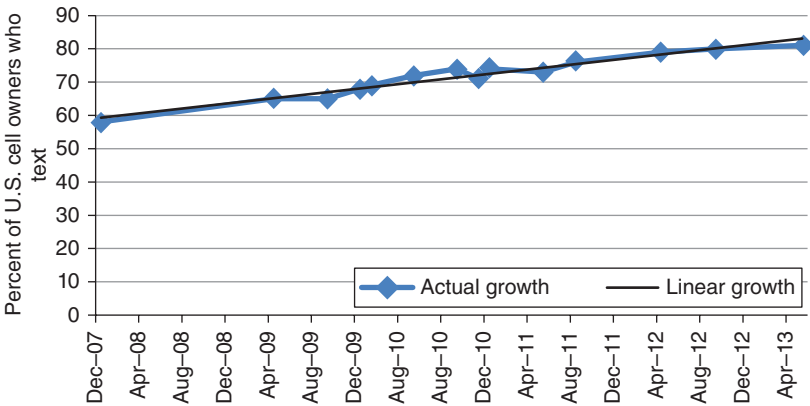


Figure 14.1 Texting rates among U.S. adults based on Pew telephone surveys from December 2007 through April 2013. Data from Pew Research Center (2013).

14.3.2 Nonresponse Error

Another source of bias would arise if sample members who do not respond to or complete text interviews—nonrespondents—differ systematically from respondents in the attributes (opinions or behaviors) measured in the survey. To our knowledge, there is almost no empirical evidence on nonresponse bias in text interviews but there is some evidence, albeit sparse and inconsistent, about response *rates* in text interviews, and it comes primarily from opt-in panels and convenience samples rather than from probability samples.

An early feasibility study found a texting response rate of 63% among online panelists who had agreed to be interviewed via SMS and had been alerted that an upcoming invitation would be sent to them (Cooke et al., 2003). In panels of mobile phone users who have agreed through their mobile service provider to be invited to participate in an SMS text interview at some future point (but who have not been explicitly prenotified about an imminent invitation) response rates reportedly vary according to country, region, and incentive (Bena, 2014), but they seem to be quite low relative to other interviewing modes; in one study using this approach with 2,046 respondents in the final sample (Schober et al., under review) (described below), the response rate was 3%.

The one direct mode comparison that we know of (Conrad et al., 2014a) suggests that response rates may be higher for text than voice interviews, at least among iPhone users who had already agreed to participate in a survey on their iPhone for a \$20 incentive. Respondents were randomly assigned to participate in one of four kinds of interviews: those conducted via text or voice by either human or automated interviewers (see Section 14.3.3 and Schober et al., 2015, for details). Response rates for text were higher with both human interviewers (72% in text vs. 52% in voice) and automated interviewing systems (49% vs. 39%) (Conrad et al., 2014a). In this dataset, although we have no evidence about how the nonrespondents might have answered the survey questions, we know some of the demographic characteristics of the nonrespondents (i.e., those who screened into the study but did not accept the invitation to participate); the nonrespondents in text did not differ from the nonrespondents in voice in age, gender, education, or race. Of course, the nonrespondents in the two modes might still differ in ways not captured in this dataset.

Another measure of nonresponse is completion rates, that is, the proportion of respondents who complete an interview (do not break off) after starting it. An early test with text interviews of three, four, and five questions (Cooke et al., 2003) showed completion rates of 99%, 93%, and 92%, respectively, suggesting that breakoff increases with more questions in a text interview. With a much longer text survey of 32 questions (Conrad et al., 2014a; Schober et al., 2015), a nonsignificant difference in completion rates is reported between text (89%) and voice (92%). This suggests that the asynchrony of text interviews that allows, or even promotes, long intervals between discrete responses (as opposed to the continuous interaction in a voice interview) does not increase the likelihood of breakoff.

As with response rates, we know the demographic characteristics of those who broke off in text and voice interviews in this study (Conrad et al., 2014a; Schober et al., 2015). Conditional on starting the interview, females were more likely to break off from text interviews than from voice interviews, which raises the possibility that breakoffs could have differential effects on nonresponse error in the two interview modes. The fact that there were no other differences by mode in the available demographic characteristics of the breakoff cases, and that overall breakoff rates did not differ in text and voice, mitigates this concern. It is also possible that in text interviews those who break off may differ from those who do not break off in ways beyond their demographic characteristics that could, in principle, be relevant to what is being measured. In any case, it remains to be seen whether these completion rate findings will generalize to other

text interview surveys, in particular those involving random and representative samples with different levels and kinds of incentives. Moreover, receiving text invitations on multiple devices (e.g., iPhone, iPad, Macbook, iMac) and the possibility of participating on one or more of several devices may affect response and completion rates (for better or worse) in text surveys.

Beyond response and completion rates, another potentially relevant issue for nonresponse error in text interviews is how sample members feel about texting as a mode of communication in general and, more specifically, as a mode of being interviewed in a survey. It is well known that demographic subgroups differ in their use of text (Duggan, 2013), and so including or not including text as an option in a survey may already lead to differential nonresponse. The participants in Schober et al.'s (2015) study reported greater willingness to participate in a future text than voice interview on their iPhone. This underscores the possibility that, at least in some populations and for some kinds of surveys, texting may well result in high participation rates, but what this means for nonresponse bias is still an open question.

14.3.3 Measurement Error: Conscientious Responding and Disclosure in Texting Interviews

Beyond coverage and nonresponse error, another major set of questions about text interviewing concerns the quality of responses in text compared to other modes. We have carried out a study (Schober et al., 2015) to compare the quality of responses in text and voice interviews on smartphones, in interviews conducted by either human interviewers or automated interviewing systems. Manipulating these two factors (interviewing medium—text vs. voice—and interviewing agent—human vs. automated) led to four experimental conditions: (i) human voice, (ii) human text, (iii) automated voice,² and (iv) automated text. Details about the modes and administration of the interviews are reported in Johnston et al. (2013) and Schober et al. (2015). In this study (from which response rates were reported in Section 3.2), 634 users of iPhones with U.S. area codes recruited through online sources (Antoun et al., 2016) volunteered to participate in a study in exchange for a \$20 iTunes gift code. Respondents were randomly assigned to one of these four modes, in which they answered 32 questions from major U.S. social surveys and methodological studies. This allowed a clean comparison of how these modes would affect responding by holding constant the mobile device and interface involved, and using native apps (phone and messages) that iPhone users use regularly for one-to-one communication, as opposed to specialized interviewing apps that would need to be downloaded and installed.

Comparing responses in voice and text interviews allowed us to test how the differences in voice and text communication outlined in Table 14.1 might affect the quality of survey responses. Comparing responses to human interviewers and automated interviewing systems allowed us to test whether the frequently observed increase in disclosure of stigmatized or compromising behavior in self-administered compared to interviewer-administered surveys (e.g., Aquilino, 1994; Tourangeau and Smith, 1996; Tourangeau and Yan, 2007) replicates on a mobile device, and whether such effects are observable in text interviews, where the social presence of an interviewer is so limited.

We measured the quality of responses in three ways. To measure conscientious responding, we computed the percentage of answers to eight questions requiring numerical responses which could be classified as either rounded (divisible by 5) or “precise” (unrounded) numbers (e.g., number of movies seen last month, number of apps on iPhone). We measured rounding as an indication of response quality because of evidence that rounded or “heaped” answers to

2 The automated voice condition was a speech interactive voice response (IVR) system, which recognized spoken answers and other speech, rather than the touchtone IVR more typical in automated telephone surveys (Bloom, 2008).

behavioral frequency questions are less likely to be accurate (Conrad et al., 1998; Holbrook et al., 2014). We also measured responses to a battery of seven questions that all used the same response scale (from “strongly favor” to “strongly oppose”), so that we could measure nondifferentiation or “straightlining,” that is, the practice of selecting the same response options for all or most questions in a battery, which is generally seen as evidence of nonconscientious responding (Krosnick, 1991). To measure disclosure, we examined responses to 15 questions with more and less socially desirable answers (e.g., number of sex partners, lifetime marijuana use, frequency of exercising, etc.), because of evidence that greater disclosure of embarrassing information is more likely to be accurate (Kreuter et al., 2008).

The findings were clear. Text interviews produced better quality data than voice interviews: significantly fewer rounded numerical answers, significantly less straightlining, and significantly more disclosure of sensitive information (see Figure 14.2). There was also an effect of

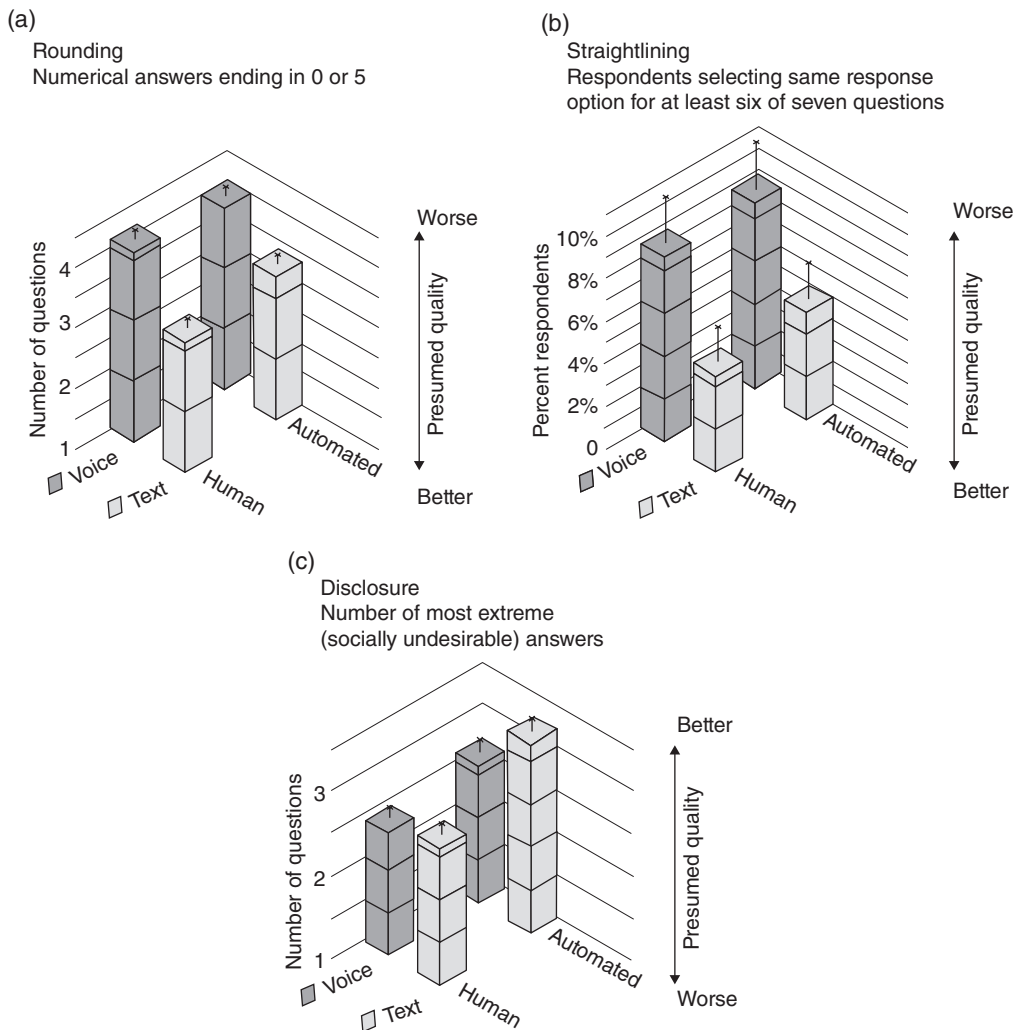


Figure 14.2 Data quality in text versus voice and human versus automated interviews; (a) rounding, (b) straightlining, and (c) disclosure. Source: Schober et al. (2015). Open access.

automation, with significantly greater disclosure of sensitive information in automated than human-administered interviews; this effect did not interact with the interviewing medium (text vs. voice) suggesting that even with few cues that a text interview is automated or interviewer-administered, automation increases disclosure as much in text as in voice interviews where the type of agent is more obvious. The improved data quality in text interviews occurred despite the fact that text respondents reported more multitasking during the course of the interview (communicating with someone else, using their iPhone to do something else, and participating in other activities). As for respondents' subjective experience of the interview, at least as many text respondents reported being satisfied as voice respondents, despite the fact that text interviews lasted longer (see Section 14.4).

What explains this pattern of results? We attribute the reduction in rounding and straightlining to the reduced time pressure of texting. Being able to respond when they are ready, and being able to carry out other tasks, seems to allow text respondents to answer thoughtfully, and potentially even to consult records to inform their answers (e.g., in answering a question about how many songs respondents have on their iPhone, to actually check the music app on their phone). We attribute the increased disclosure in texting (consistent with results reported by West et al., 2015) to the reduced social presence of either the human interviewer or the automated system: text is asynchronous and there is less immediate evidence of an interviewer's reaction to answers. The lack of pressure to respond quickly in text may allow respondents to grow comfortable with the potential consequences of disclosing embarrassing information, and texting an answer to a sensitive question might feel more "private" than speaking it out loud.

Might different rates of nonresponse account for these findings? That is, were more conscientious sample members (i.e., those less likely to round or straightline) or those more willing to disclose sensitive information *less* likely to start voice than text interviews? This is an unlikely explanation for the pattern observed: our sample of volunteers had all agreed to participate in an iPhone survey before they were randomly assigned to an interviewing mode, and so motivation should not have differed across the four conditions. As described in Section 14.3.2, it is also unlikely that these results are due to nonresponse error, as there was no evidence that the demographic characteristics of nonrespondents differed between text and voice surveys.

The findings from this study might be surprising if one thinks of texting as a mode of interaction in which people are casual and not particularly careful, or if one thinks of texting as a mode of abbreviated communication (e.g., "LOL" for "laugh out loud" or "IMHO" for "in my humble opinion") developed to reduce input effort and conserve characters. They might also be surprising if one imagines that respondents might worry about leaving a visible and persistent trace of their embarrassing answers that others might see. The evidence from this study suggests quite the opposite; measurement error in text interviews may well affect estimates less than in spoken interviews, at least in text and voice interviews as implemented here.

14.3.4 Measurement Error: Interface Design for Texting Interviews

Another set of questions about text interviewing concerns how response quality will be affected by the interface that respondents use: the display of interview dialog (e.g., threaded questions and answers vs. single message displays), the input interface (physical numeric vs. virtual alphanumeric keyboard), and the kinds of responses that are acceptable (e.g., can the interviewer or automated system recognize and accurately classify mis-spelled or open-ended responses?). Even to design an interface for the broadest set of mobile devices (e.g., one that works on the smallest screens and within the constraints of SMS), there is still a range of design considerations. Organizations that deploy text interviews clearly have made choices about interface

design, but little systematic methodological research on best practices for text interviewing interfaces has yet been carried out.

We have conducted one study (Schober et al., under review) to test two hypotheses about the effects of requiring single-character responses (e.g., “y” or “n”) rather than multicharacter responses (e.g., “yes” or “no”) on the distribution of responses in text interviews. The first hypothesis is about respondent effort, based on the fact that creating text messages usually involves at least one motor action per texted character—one action per character with virtual alphanumeric keypads, even more with physical numerical keypads. Effort to enter answers thus increases as respondents enter more characters. To the extent that respondents try to minimize effort, they may select shorter response options (i.e., fewer characters) in a text interview that requires multiple character responses, even if the shorter response option does not fit their circumstances as well as a longer one. Our *effort hypothesis* is thus that text survey respondents will be more likely to choose responses that require fewer keystrokes, independent of answer content.

Second, we tested a *sensitivity hypothesis*: text survey respondents may be more likely to select more sensitive response options when they are single character—that is, with arbitrary rather than meaningful and thus potentially value-laden labels—than when respondents must explicitly report (type) the full answer. So, for example, when answering questions about whether one has engaged in risky sexual behavior it might be easier to “hide behind” a single-character response like “1” than typing out “yes” or even typing the character “y,” let alone keying in full descriptions of risky behaviors.

The study involved 2,046 French-speaking respondents, recruited from GeoPoll’s Tunisian mobile sampling frame of over two million mobile subscribers, who had agreed to be contacted to participate in text messaging interviews in exchange for a credit on their mobile phone bills. Respondents texted answers to 12 closed-form survey questions that were texted to them by an automated system (all in French). Half the questions were on topics we judged to be sensitive (e.g., “Has your household paid a bribe to any of the following in the past six months: university, government official, traffic police, tax official? Yes or No”) and half were on topics less likely to be sensitive (e.g., “During the past week, have you eaten in a restaurant? Yes or No”). Half the questions were yes–no questions like these, and half were questions with several response categories, with answers of different character lengths (e.g., “How often do you attend religious services: At least once a week, almost every week, about once a month, seldom, or never?”).

Respondents were randomly assigned to either a single- or multicharacter response condition; for the yes–no questions, respondents in the single-character condition would enter “1” for “oui” (“yes”) or “2” for “non” (“no”), while in the multicharacter condition they would need to enter the entire word “oui” or “non.” For the questions with several response options, the difference in character length of the response options could be large, e.g., entering “1” for “au moins une fois par semaine (at least once a week)” would save a particularly large number of characters and, therefore, effort.

The findings on the effort hypothesis were clear. Overall, respondents in the multicharacter condition selected the shortest response option significantly more often, and the longest response option significantly less often, than respondents in the single-character condition. For example, the percentage of respondents reporting the *shortest* response option (*lire*, i.e., reading) as their favorite recreational activity differs between single- and multicharacter conditions (9% vs. 17%), as does the proportion reporting the *longest* response option, chatting with friends (*discuter avec des amis*), as their favorite activity (36% vs. 24%).

As for the sensitivity hypothesis, respondents in the single-character condition reported more socially undesirable answers than respondents in the multicharacter condition, considering (together) the five sensitive questions for which we could identify the less socially

desirable answers (crime victimization, bribery, voting, drinking, and religious attendance). This effect cannot result only from differences in the relative effort of typing the different response options; respondents were significantly more likely to report having paid a bribe with a single-character response (“1”) than by typing out “oui” (“yes”) versus “non” (“no”), which both require entering three characters. This example suggests that there may well be a benefit in text interviews to allowing respondents to “hide behind” a single, nonmeaningful character for their sensitive answers.

These findings demonstrate that, although the design considerations for text survey interfaces are different than those in other modes, responses can be shaped by design choices much as they can be in spoken interviews and other textual modes like browser-based surveys. Much more systematic research will be needed to discover how interface choices affect measurement error, especially as texting interfaces, devices, and practices evolve.

14.4 Costs and Efficiency of Text Interviewing

The costs and efficiency of text surveys relative to other modes are likely to be a determining factor for how text surveys fit into the suite of available data collection methods. If text surveys are less efficient than other modes, then researchers must weigh the tradeoffs between the kinds of improvements to response quality observed thus far (Schober et al., 2015; West et al., 2015) and increased survey costs. On the other hand, if text surveys are no less efficient—or more efficient—than other modes, then the potential data quality advantages of text are particularly attractive.

One way to assess the efficiency of text surveys is by examining response and completion rates in text compared to other interviewing modes, as discussed in Section 14.3.2. Important variables are likely to include the extent to which sample members know that they might be invited to participate in a text interview (e.g., whether they have screened in to a study), and the extent to which they have previously indicated their commitment to participating in surveys more generally. The fact that “cold” (unsolicited) texts are, in many parts of the world, not permissible amplifies the importance of prior consent to be contacted and to participate for text interviewing, which in turn becomes a cost consideration for researchers.

Beyond response and completion rates, another dimension of efficiency is timeliness of data collection, which can be measured in a few ways. First, there is the simple total amount of time an interview takes. Early evidence suggested that text interviews may lead to surprisingly speedy completion; Cooke et al. reported receiving 36% of their responses to 3–5 texted questions in less than one minute and 58% within two minutes; 82% were received within an hour. In Schober et al.’s (2015) study, which involved more (32) questions, text interviews lasted significantly longer than voice interviews, but they involved many fewer conversational “turns”—messages in the text interviews—suggesting that text communication was spread out over a longer period of time but was more succinct than corresponding voice communication (see Figure 14.3). Another measure of efficiency is the amount of time a case is in the field from initial contact to completion. Conrad et al. (2014a) report that text interviews—especially automated text—were completed more quickly after the invitation was sent than voice interviews (with a median time in field of 0.5 hours in automated text, 1.1 hours in human text, 4.5 hours in automated voice, and 10.8 hours in human voice). They attribute the shorter field periods for text interviews to the reduced number of contact attempts required to obtain an interview relative to voice interviews (see Section 14.2). Text also led to a larger proportion of the sample that completed the survey within one day of receiving

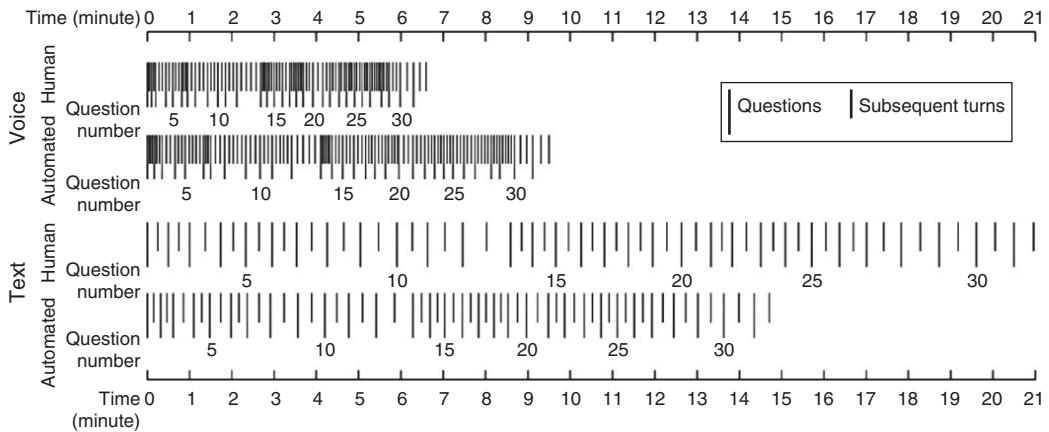


Figure 14.3 Interview duration (median duration of question–answer sequences in minutes) and median number of turns per survey question in text and voice interviews. Source: Schober et al. (2015). Open access.

an invitation (93% in automated text, 71% in human text, 59% in automated voice, and 51% in human voice). So the overall pattern suggests that text interview administration can lead to rapid data collection, particularly when automated.

While, of course, in automated interviews there are no personnel costs associated with each additional interview, when human interviewers administer text interviews another measure of efficiency is interviewer productivity, measured as the amount of interviewer time spent obtaining and conducting interviews where less time reflects greater productivity. In Conrad et al.'s (2014a) analyses, interviewers in text spent more overall (elapsed) time per interview than they did in voice (1.9 hours vs. 0.8 hours). This was further divided into time spent recruiting participants even if waiting for a response (text: 1.4 hours per completed interview; voice: 0.7 hours per completed interview) and time spent interviewing them as measured by median interview length (text: 30 minutes; voice: 8 minutes). As implemented in this study, text interviews led to lower interviewer productivity than voice. But this is likely due to the two artificial constraints in the study: allowing interviewers to conduct only one text interview at a time (for comparability with the voice interviews, for which only one interview can be conducted at a time), and instructing interviewers to wait for a text response for 15 minutes after texting an invitation or a question (which is why the field periods for text interviews are shorter but interviewers are less efficient administering text than voice interviews, i.e., they spend more time waiting in this implementation). In production implementations, interviewers could easily conduct more than one text interview at once, and protocols for how long to wait before moving on would likely be different (shorter), both of which would almost certainly increase the interviewer productivity in text interviews.

The general picture thus far, then, is that text interviews are quite efficient by most measures, and that the benefits to measurement through text seem not to be accompanied by additional costs, and if anything they are less costly. Of course, any assessments of the efficiency and costs of text interviews must be considered in light of the extent to which respondents have committed to participating in a future text interview, and the laws surrounding unsolicited text messages. Efficiencies are also likely to differ between automated and human-administered text interviews, as well as in different populations of users answering different kinds and numbers of questions on different devices.

14.5 Discussion

The findings reviewed here suggest that text interviewing is a viable mode both for contacting and interviewing members of the public. Based on the empirical evidence to date, the characteristics of text users are more or less the same as those of mobile phone users who do not text, and so coverage of the mobile phone population does not seem to be an obstacle to generalizing results from text interviews to (at least) that population. Text messaging seems to be an effective mode for contacting potential participants, at least those who have agreed to be contacted via text. Text interviews can produce data whose quality is as good as, or in many cases better than, the quality of data from voice interviews: more precise numerical answers (less rounding) and less frequent straightlining (Schober et al., 2015), and greater disclosure of sensitive information (Schober et al., 2015; West et al., 2015). Thus, text interviews seem to be cost-effective, particularly when automated. Given the growing use of texting in daily communication globally, text interviewing should be considered more broadly as a tool for social measurement.

But much more needs to be known about when and how text contacts and interviews are effective: will the reductions in measurement error described here be observed in samples with more and less experience or comfort with texting, in samples who have not already agreed to be contacted or interviewed via text, or on different devices with different texting capabilities? Is texting suitable for longer interviews, or only for brief ones, and what incentives would be required for respondents to participate in longer interviews? The technologies for texting also continue to evolve: for example, the ability to dictate text messages by speaking and the ability to hear text-to-speech output from incoming messages could allow the hundreds of millions of users who do not read or are visually impaired to use the medium. This could be particularly relevant to coverage error in countries with low literacy rates. The ability to attach audio or video to texts (Fuchs, 2008) could become commonplace for at least some populations. How will the evolution of texting technologies affect TSE?

In the near term, one can guard against coverage error due to the range of texting devices and interfaces by designing for the broadest set of devices and interfaces: within the constraints of SMS texting on small screens, without relying on a threaded display of messages or assuming that users have a virtual keypad. This further argues for single-character responding if at all possible, considering the higher levels of motor effort required to enter text on numeric keypads. Designing an interview in this way strips away all but the essential components: questions and response options, with no visual bells and whistles or multimedia extras (like radio buttons for response options, sliders, progress indicators, or color) that can affect measurement error in other modes, particularly web surveys (Tourangeau et al., 2013). In a sense, an interview designed in this way implements a visual analog of task-oriented “no-nonsense” spoken interaction, without the intonation and politeness cues inherent in spoken interactions (and in which researchers usually do not intend to play a role (Conrad et al., 2014b)). This may be a positive or negative aspect of text interviews; the stripped-down character might be part of what reduces socially desirable responding, but it might also remove the motivational benefits of interacting with a human interviewer.

How researchers include text interviewing in the suite of modes they implement—for example, in a mixed-mode survey that allows texting as one option—raises new questions about what counts as comparability across modes. On the surface, text interviews look so minimalist that they may not seem comparable to other more richly formatted textual modes of questioning, let alone spoken or face-to-face data collection. We propose that what matters more (or at least as much) will be comparability of the respondent experience—that is, being able to interpret questions and provide responses in a way that respondents find as convenient and comfortable, promoting the same level of effort and openness, as in other modes. As we see it, the issue of

cross-mode comparability has become more complicated; the fact that text interviews can lead to greater disclosure and more conscientious responding than voice interviews actually raises the question of whether telephone or face-to-face interviewing should still be considered the gold standard of survey quality, to which all other modes are compared.

In any case, survey researchers will need to grapple with the fact that so many potential survey respondents use text messaging as a central part of their daily communication, and not only for casual personal purposes. *Not* including text messaging in a study's design may well require an active choice as texting becomes more ubiquitous.

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